1. Introduction

Attrition refers to leaving of employees from an organization due to reasons like turnover, resignation, and retirement. Attrition rate is the percentage of employees who leave among total employees in a given time period. Attrition is one of the most serious issues in organizations since it directly impacts productivity, HR costs and smooth flow of work. It is necessary to understand the causes of attrition in order to address it properly.

In this project, I analyzed the problem of attrition in an organization. The focus is on finding the factors affecting attrition among employees and providing actionable insights to HR for addressing the attrition problem.

Data analytics methods, like Exploratory Data Analysis (EDA) and Data Visualization were used to analyze the factors like income, overtime and age in the dataset. SQL was used for data querying and analysis, and Power BI was used to create an interactive dashboard to give a summary of the insights visually.

2. Objectives

The primary objective of this project is to analyze employee data to obtain meaningful insights regarding the attrition. The specific objectives of the study are as follows:

a. Identify key factors influencing attrition, including overtime, salary, job satisfaction, and work-life balance
b. Develop an interactive dashboard using Power BI to visualize attrition metrics and insights
c. Generate actionable insights and recommendations to support data-driven HR decision-making

3. Methodology

This analysis follows a quantitative approach to analyze employee attrition using the *IBM HR Analytics Employee Attrition & Performance* dataset sourced from the Kaggle platform. The dataset includes variables such as age, job role, monthly income, job satisfaction, and attrition status.
The data was already cleaned, so no further cleaning was needed. Exploratory data analysis was conducted using **SQL** (using group by, case when, and aggregations, etc) to identify patterns and relationships across features.
Insights were further visualized through an interactive dashboard built in **Power BI**, leveraging descriptive statistics and visual exploration techniques to uncover key factors influencing employee attrition. Slicers were used for dynamic analysis.

4. Analysis and Interpretation

4.1. Overview of the dataset

The dataset consists of a total of **1,470 employee records** and includes **multiple variables** representing demographic details, job-related characteristics, and employee behavior. These variables provide a comprehensive view of the workforce and enable effective analysis of attrition patterns..
Key features of the dataset include employee age, job role, monthly income, job satisfaction, work-life balance, years of experience, and attrition status. The dataset consists of structured data with both numerical and categorical variables, enabling comprehensive analysis.

The target variable in this study is **employee attrition**, which indicates whether an employee has left the organization. The dataset provides a suitable foundation for identifying trends, relationships, and key factors influencing attrition.

4.2. **Exploratory Data Analysis**

Exploratory Data Analysis (EDA) was conducted to examine patterns and relationships within the dataset and to identify key factors influencing employee attrition. Various visualizations and summary statistics were used to interpret the data effectively.

The analysis shows that a significant proportion of employees have not left the organization; however, a noticeable percentage of attrition exists, indicating potential underlying issues.

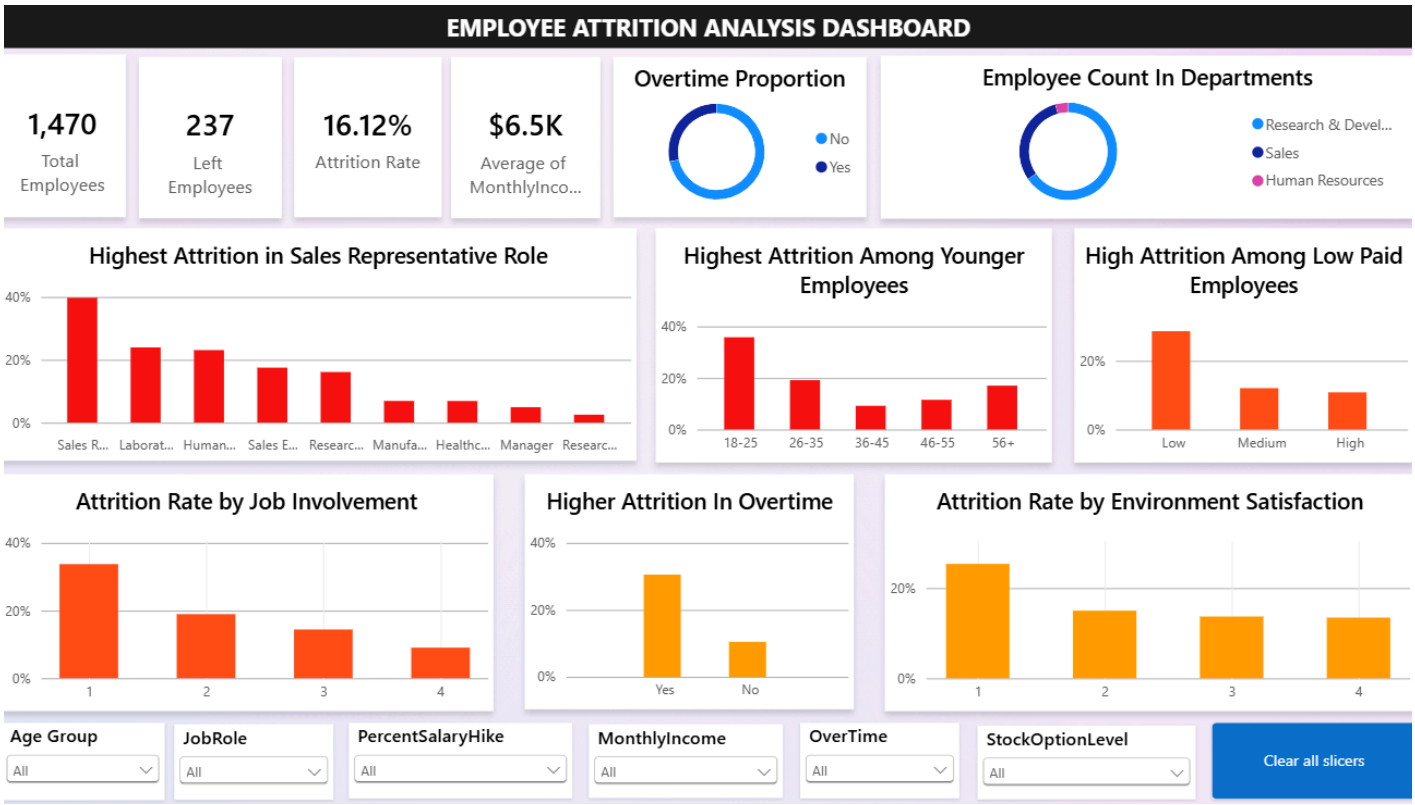


Figure 1. Employee Attrition Analysis Dashboard

In the dashboard, the trends strongly associated with attrition are shown. These include particular job roles, younger employees, low paid employees, employees with poor job involvement, employees working overtime, and employees with poor environment satisfaction.

4.2.1. **Attrition by Job Role**

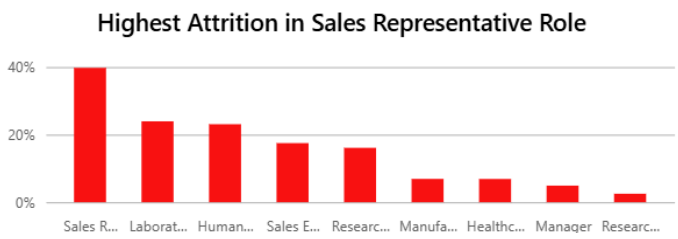


Figure 2. High Attrition Job Roles

As shown in the chart, attrition rates vary across different job roles within the organization, with the highest observed in Sales Representative (39%), followed by Laboratory Technician and Human Resources (23%). Roles such as Sales Executive and Laboratory Technician show relatively higher attrition, possibly due to factors like workload, performance pressure, and limited growth opportunities. In contrast, roles like Research

Scientist and Manager exhibit lower attrition, indicating better job satisfaction and stability.

Overall, the analysis highlights that job role-specific factors significantly influence attrition, and the organization should adopt targeted retention strategies accordingly.

4.2.2. **Attrition by Age Group**

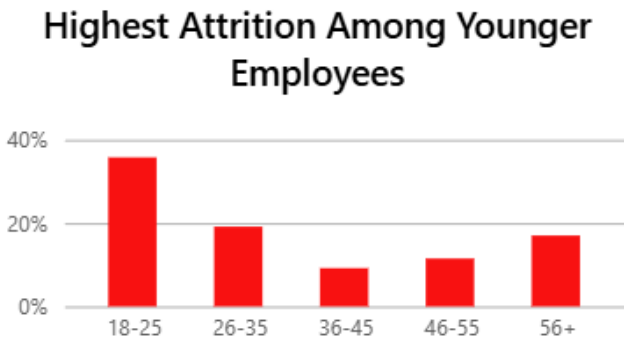


Figure 3. High Attrition Among Young Employees

As shown in the chart, attrition is highest among employees aged 18–25 (35.7%), followed by the 26–35 group (19.2%). The rate declines across middle age groups but shows a slight increase again among senior employees, although it remains lower than the youngest group.

This pattern indicates that while younger employees are the most likely to leave, attrition is not strictly linear across age groups. The findings highlight the need for targeted retention strategies, particularly for early-career employees, while also monitoring attrition trends among senior workforce segments.

4.2.3. **Attrition Rate by Salary Slab**



Figure 4. Low Paid Employees Quit More

From the chart, it can be understood that a clear inverse relationship with salary levels exists. Employees in the low salary slab exhibit the highest attrition rate (28.6%), followed by those in the medium and high salary groups.

This pattern indicates that as salary increases, attrition tends to decline. The findings highlight compensation as a key factor influencing employee retention, suggesting the need for competitive pay structures and targeted retention strategies for lower-income employee groups.

4.2.4. **Attrition by Job Involvement**

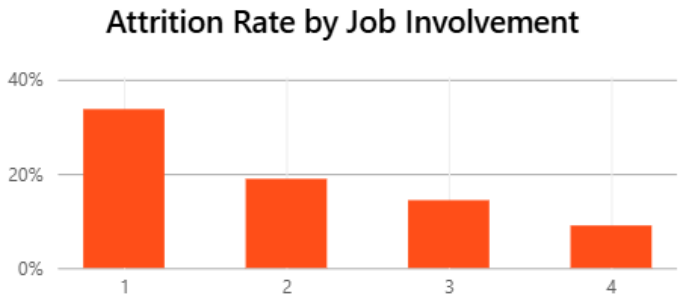


Figure 5: High Attrition Among Low Job Involvement Employees

As shown in the chart, attrition rates are highest among employees with low job involvement, with the peak observed at involvement level 1 (33.7%), followed by level 2 (18.9%). Attrition declines as job involvement increases, indicating stronger retention among highly engaged employees.

This pattern suggests that employee engagement plays a critical role in attrition. The findings highlight the importance of improving job involvement through meaningful work, recognition, and growth opportunities to enhance retention.

4.2.5. **Attrition by Overtime**

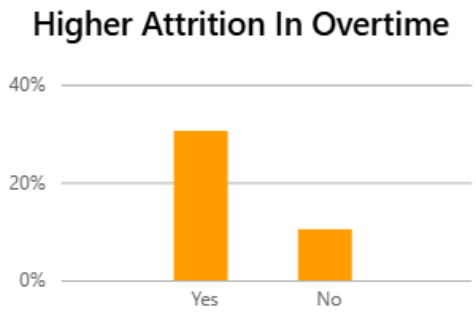


Figure 6. Employees Working Overtime Leave More

The chart shows that attrition is significantly higher among employees working overtime (30.5%) compared to those who do not (10.4%).

This substantial gap indicates a strong association between overtime and employee attrition. The findings highlight the importance of managing workload and maintaining work-life balance to improve employee retention.

4.2.6. **Attrition by Environment Satisfaction**

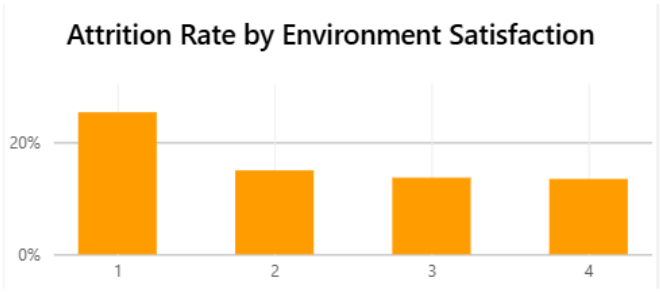


Figure 7. Employees with Poor Environment Satisfaction Leave More

As shown in the chart, attrition is highest among employees with the lowest environment satisfaction rating (1), at 25.3%. Attrition decreases as satisfaction levels increase, indicating stronger retention among employees with more favorable work environments.

This pattern suggests a strong relationship between environment satisfaction and attrition. The findings highlight the importance of maintaining a positive work environment to improve employee retention.

5. Key Insights

- **Overtime Impact:** Employees working overtime show significantly higher attrition (30.5%) compared to those who do not (10.4%), indicating workload as a major risk factor.
- **Compensation Effect:** Attrition is highest among lower salary groups (28.6%) and decreases as salary increases, highlighting compensation as a key retention driver.
- **Job Role Variation:** Roles such as Sales Representative, Laboratory Technician, and Human Resources exhibit higher attrition, suggesting role-specific challenges.
- **Employee Engagement:** Lower job satisfaction and involvement levels are associated with higher attrition, emphasizing the importance of engagement.
- **Work-Life Balance:** Poor work-life balance correlates with increased attrition, reinforcing the need for balanced work policies.
- **Age Factor:** Younger employees (18–25) show the highest attrition, with a decline in mid-career stages and a slight rise again among senior employees.
- **Business Travel:** Frequent business travel is linked to higher attrition compared to rare or no travel.
- **Tenure & Department:** Attrition is higher among employees with shorter tenure and varies across departments, indicating early-stage and structural retention challenges.

6. Recommendations

- **Optimize Workload & Overtime:** Employees working overtime show significantly higher attrition. Implement better workload

distribution, realistic deadlines, and policies to minimize excessive working hours.

- **Ensure Competitive Compensation:**
  Higher attrition among lower salary groups highlights the need for fair and competitive pay structures aligned with industry standards and performance.
- **Improve Work-Life Balance:**
  Introduce flexible work arrangements, remote work options, and wellness initiatives to support employee well-being and reduce burnout wherever possible.
- **Enhance Employee Engagement:**
  Strengthen job satisfaction through recognition programs, career development opportunities, and a positive work environment.
- **Manage Business Travel Impact:**
  Provide support for travel-intensive roles through flexible policies, adequate rest periods, and additional benefits.
- **Focus on Early Tenure Retention:**
  Improve onboarding, mentorship programs, and career path clarity to retain employees in the early stages of their tenure.
- **Adopt Department-Specific Strategies:**
  Tailor retention strategies based on department-level challenges identified through analysis.
- **Leverage HR Analytics:**
  Continuously monitor attrition metrics and use data-driven insights to take proactive retention measures.

Summary:
A holistic approach combining workload management, fair compensation, employee engagement, and data-driven decision-making is essential to reduce attrition and build a stable workforce.

7. Conclusion

This project analyzed employee attrition using the *IBM HR Analytics Employee Attrition & Performance* dataset to identify key factors influencing employee turnover. The analysis shows that attrition is driven by a combination of factors, with overtime, compensation, job satisfaction, and work-life balance emerging as the most significant.

The findings highlight the importance of data-driven HR strategies, enabling organizations to improve retention through better workload management, competitive compensation, and enhanced employee engagement. Overall, the project demonstrates how analytics can be used to uncover actionable insights and support effective decision-making in workforce management.

8. Limitations

- **Use of Fictional Data:**
  The analysis is based on a dataset created for learning purposes and does not include real world organizational data, limiting direct insights into employee perspectives.
- **Missing Variables:**
  Key factors such as organizational culture, leadership style, and employee relationships are not included, which may influence attrition but are not captured in the analysis.
- **Correlation, Not Causation:**
  The analysis identifies relationships between variables and attrition but does not establish causal effects.
- **No Financial Impact Analysis:**
  The dataset does not include cost-related metrics due to lack of relevant data, limiting insights into the financial implications of employee attrition.